Marmara Üniversitesi
Mühendislik Fakültesi
Elektrik-Elektronik Mühendisliği

Laboratory Manual

EE 3011
Electronics I

Experiment (4) BJT
IV Characteristics

Spring 2023

Experiment (4)
BJT IV Characteristics

Objectives

- To study the IV curve of a BJT.
- To experimentally evaluate the DC gain parameter of a BJT.
- To study the transfer function of a BJT circuit.

Pre-lab Work

Analyze the DC circuit below. Assume the transistor is in active mode and find expressions for the collector current and the collector-emitter voltage.

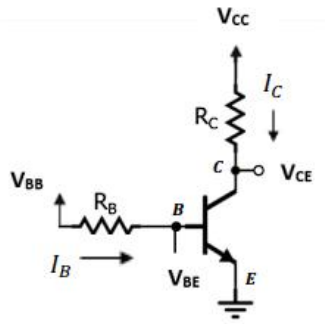


Figure 1: A BJT DC Circuit.

Solution

Equipment

Table 1. Equipment for experiment 8.

	Equipment	Model	Serial No.	Stock No.
1	BreadBoard Station			
2	Resistors 5 kW (1), 10 kW (1), 100 kW (1)			
3	BJT NPN transistor (1)	2N3904	https://www.onsemi.com/pub/Collateral/2N3903-D.PDF	
4	Multi-meter (1)			
5	Probes and cables			

Experimental Procedure

Part (1): BJT IV Characteristics

1. Connect the circuit in figure 1 with R_c=10 k and R_B=100 k. Pay a special attention to how you connect the terminals of the BJT. Refer to the data sheet to identify the transistor terminals.

2. Set V_{BB} to a constant voltage of +10 V while connecting V_{CC} to a variable DC voltage source.
3. The IV characteristics (I_C versus V_{CE}) is obtained by slowly varying V_{CC} and measuring the voltage V_{CE} and V_{RC} at each setting. Use table 2 to enter your results. Also record the measured values of V_{RB} at each setting (to be used in part 2 of the experiment).

Part (2): BJT DC Gain Parameter (β)

1. The required data have been collected in part 1.

Part (3): BJT Transfer Function

1. Reconsider the circuit in figure 1.
2. Set V_{CC} to a constant voltage of +10 V while connecting V_{BB} to a variable DC voltage source.
3. The transfer function (V_{CE} versus V_{BB}) is obtained by slowly varying V_{BB} and measuring the voltage V_{CE} at each setting. Use table 3 to enter your results.
4. Repeat step 3 for another value of R_C, say 5 k. Record your results in table 3.

Results and Analysis

Part (1): BJT IV Characteristics.

1. Calculate the currents I_C and I_B values in table 2 (Use $I_C = V_{RC}/R_C$ and $I_B = V_{RB}/R_B$).
2. Use graph 1 to sketch the IV curve.

Table 2. IV Characteristics Measurements.

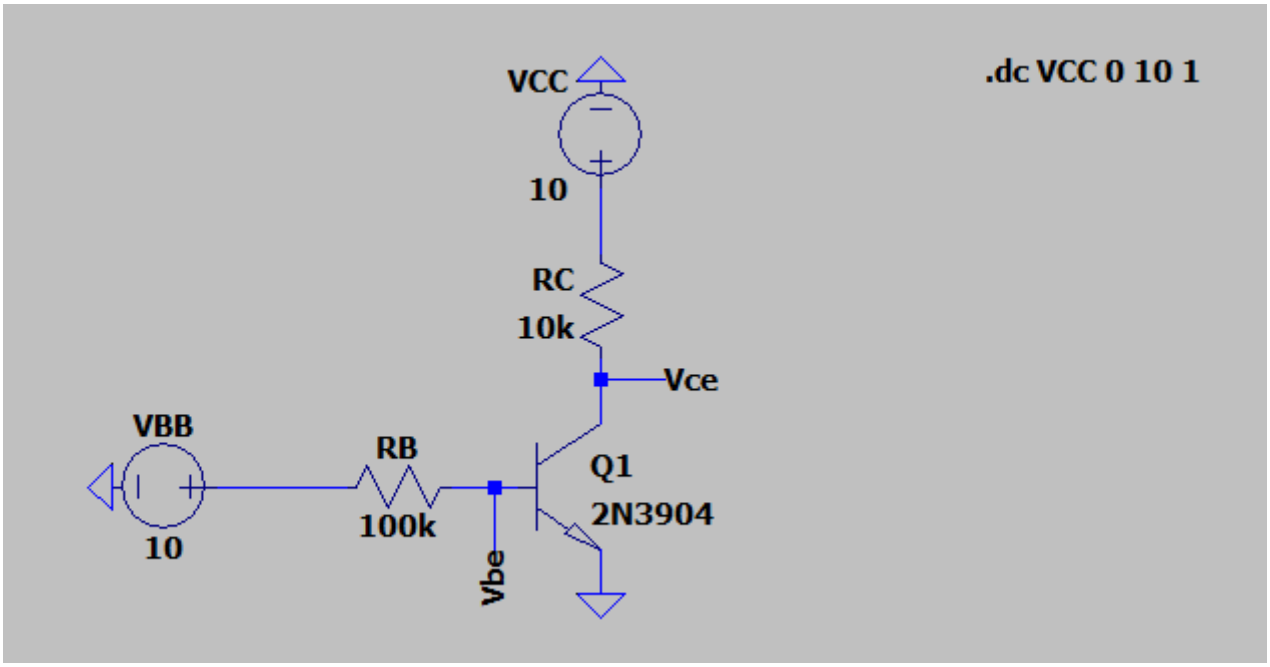
Vcc	V_{CE}(mV)	V_{RC}	I_C	V_{RB}	I_B	$\beta = I_C/I_B$
0	5,787	5,787E-06	5,787E-07	0,009363	0,00009363	0,006181
1	10,922	0,00098908	0,000098908	0,009358	0,00009358	1,056935
2	15,2456	0,00198476	0,000198476	0,0093538	0,000093538	2,121876
3	18,99	0,002981	0,0002981	0,0093502	0,000093502	3,188167
4	22,29	0,0039777	0,00039777	0,009347	0,00009347	4,25559
5	25,24	0,00497476	0,000497476	0,0093441	0,000093441	5,323958
6	27,91	0,0059721	0,00059721	0,00934	0,0000934	6,394111
7	30,36	0,00697	0,000697	0,0093392	0,000093392	7,463166
8	32,62	0,00796738	0,000796738	0,0093371	0,000093371	8,533035
9	34,71	0,0089652	0,00089652	0,0093351	0,000093351	9,603754
10	36,67	0,0099632	0,00099632	0,009333	0,00009333	10,67524

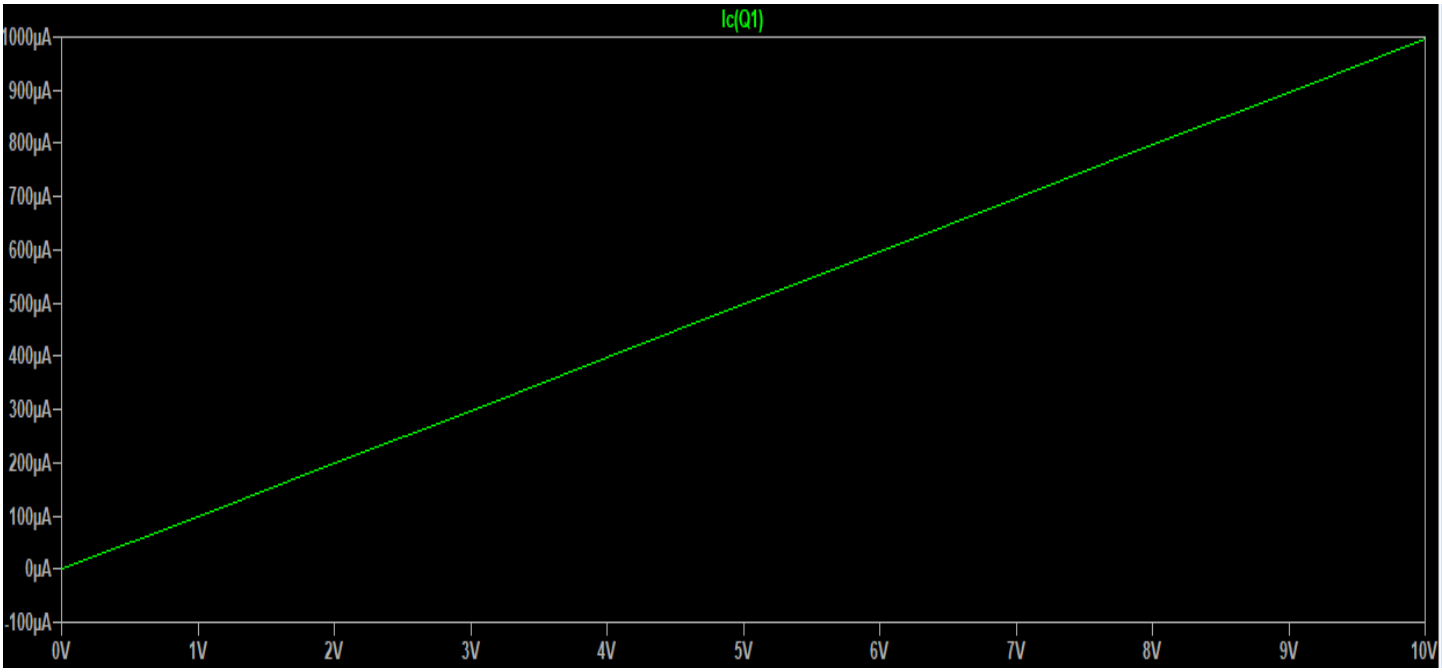
3. Comment on the IV curve that you obtained. Can you identify the transistor modes of operation? Show your answer on the curve.

Answer: The transistor is in the cutoff region.

4. Is the collector current constant in the active mode region of the IV curve? Find an approximate value of the Early voltage.

Answer: Not constant.





Graph 1: Transistor IV Curve.

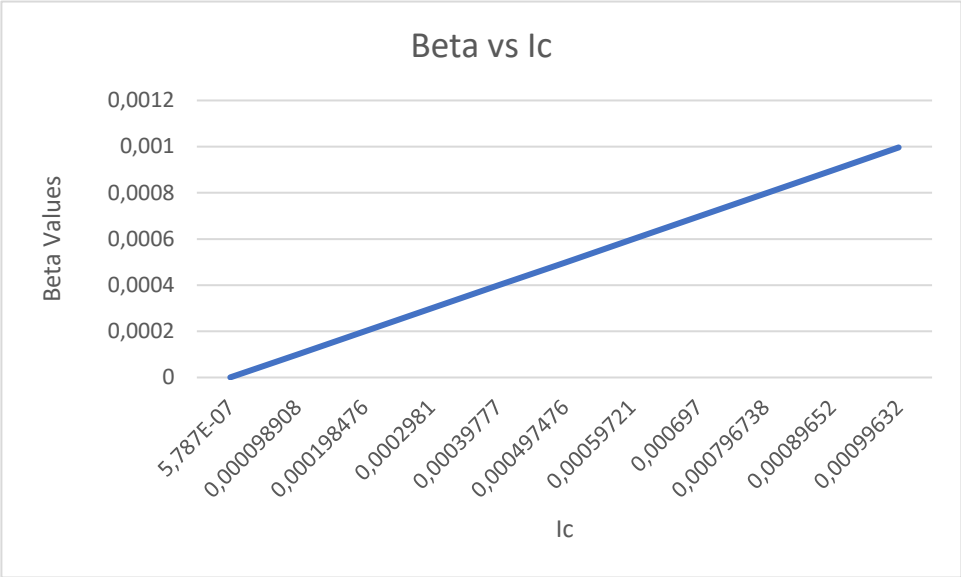
Part (2): BJT DC Gain Parameter (β)

1. Calculate the current I_B values in table 2. Also, calculate the DC gain parameter (β).
2. Why is the value of β not constant in the table? Explain in details.

Answer:

3. Use graph 2 to sketch β vs. I_C relation. Compare to the curve given in the transistor data sheet and give your comments.

Answer:



Graph 2: DC gain parameter.

Part (3): BJT Transfer Function

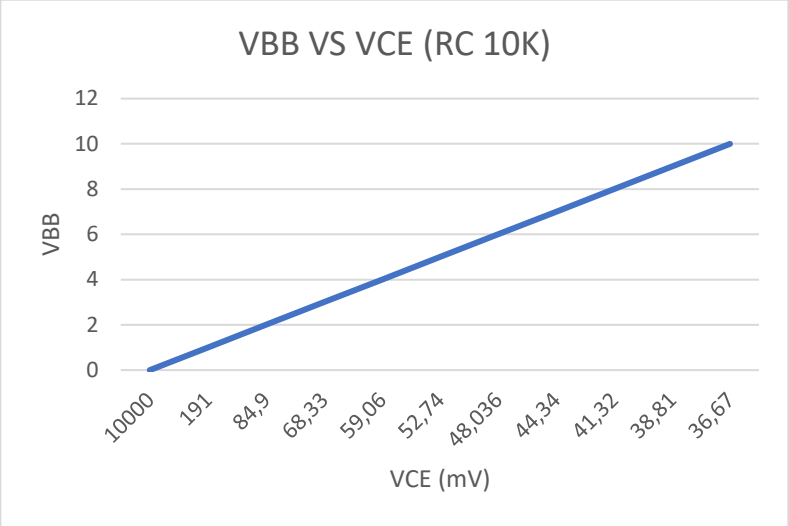
1. Use graph 3 to sketch the two transfer functions using the data in table 3, for both R_C values.
2. Can you identify the transistor modes of operation? Show your answer on the curves.

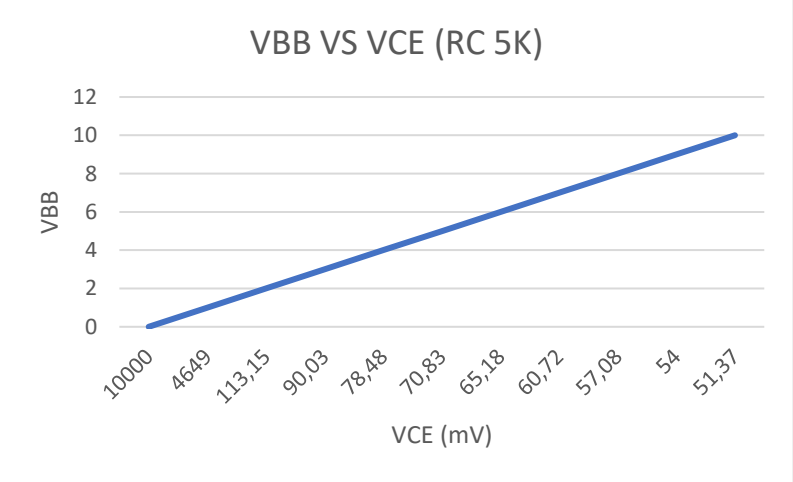
Answer: The transistor is in the cutoff region.

Table 3. Transfer Function Measurements.

	R_C 10k	R_C 5 k
V_{BB}	V_{CE} (mV)	V_{CE}
0	10000	10000
1	191	4649
2	84,9	113,15
3	68,33	90,03
4	59,06	78,48
5	52,74	70,83
6	48,036	65,18
7	44,34	60,72
8	41,32	57,08
9	38,81	54
10	36,67	51,37

Graph 3: Circuit transfer function.





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